MATH 128 - HW10 - Jie Wen

Problem 1: Problems from the textbook, chapter 3.4

2a

Let

$$x_0 = 0, \quad x_1 = 0.5.$$

The degree-1 Lagrange polynomials are

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - 0.5}{0 - 0.5} = 1 - 2x,$$
$$L_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - 0}{0.5 - 0} = 2x.$$

Derivatives at the nodes:

$$L_0'(x) = -2 \Rightarrow L_0'(x_0) = -2, \qquad L_1'(x) = 2 \Rightarrow L_1'(x_1) = 2.$$

For $n = 1$, the Hermite basis pieces are

$$H_{1,0}(x) = \big[1 - 2(x - x_0)L_0'(x_0)\big]L_0(x)^2,$$
$$H_{1,1}(x) = \big[1 - 2(x - x_1)L_1'(x_1)\big]L_1(x)^2,$$
$$\hat{H}_{1,0}(x) = (x - x_0)L_0(x)^2, \qquad \hat{H}_{1,1}(x) = (x - x_1)L_1(x)^2.$$

Plug in the values:

- Since $x_0 = 0$, $L_0'(x_0) = -2$:

$$\begin{aligned} H_{1,0}(x) &= \big[1 - 2x(-2)\big](1 - 2x)^2 \\ &= (1 + 4x)(1 - 2x)^2 \\ &= 16x^3 - 12x^2 + 1. \end{aligned}$$

- Since $x_1 = 0.5$, $L_1'(x_1) = 2$:

$$\begin{aligned} H_{1,1}(x) &= \big[1 - 2(x - 0.5) \cdot 2\big](2x)^2 \\ &= [1 - 4(x - 0.5)]4x^2 \\ &= (3 - 4x)4x^2 \\ &= -16x^3 + 12x^2. \end{aligned}$$

- The “derivative” basis:

$$\hat{H}_{1,0}(x) = x(1 - 2x)^2 = 4x^3 - 4x^2 + x,$$
$$\hat{H}_{1,1}(x) = (x - 0.5)(2x)^2 = (x - 0.5)4x^2 = 4x^3 - 2x^2.$$

The Hermite cubic is

$$H_3(x) = f(x_0)H_{1,0}(x) + f(x_1)H_{1,1}(x) + f'(x_0)\hat{H}_{1,0}(x) + f'(x_1)\hat{H}_{1,1}(x).$$
$$H_3(x) \approx 2.25376x^3 + 1.74624x^2 + 2x + 1.$$

4a

The Hermite interpolating cubic is:

$$H_3(x) = (24 - 8e)x^3 + (8e - 20)x^2 + 2x + 1.$$
$$\begin{aligned} H_3(0.43) &= (24 - 8e)(0.43)^3 + (8e - 20)(0.43)^2 + 2(0.43) + 1 \\ &\approx 2.362071014. \end{aligned}$$

The polynomial approximation is

$$f(0.43) \approx H_3(0.43) \approx 2.36207.$$
$$f(0.43) = e^{2 \cdot 0.43} = e^{0.86} \approx 2.363160694.$$

Absolute error:

$$\begin{aligned} |f(0.43) - H_3(0.43)| &\approx |2.363160694 - 2.362071014| \\ &\approx 0.00108968. \end{aligned}$$

So,

$$f(0.43) \approx 2.36207, \quad \text{absolute error} \approx 1.09 \times 10^{-3}.$$

11

(a)

Assume $P(x)$ is another polynomial of degree at most $2n+1$ that agrees with f and f' at the points $x_0, \ldots, x_n$, i.e., $P(x_i) = f(x_i)$ and $P'(x_i) = f'(x_i)$ for each $i = 0, 1, \ldots, n$.

Define $D(x) = H_{2n+1}(x) - P(x)$. Then $D(x)$ is a polynomial of degree at most $2n+1$, and

$$D(x_i) = H_{2n+1}(x_i) - P(x_i) = f(x_i) - f(x_i) = 0, \quad D'(x_i) = H'_{2n+1}(x_i) - P'(x_i) = f'(x_i) - f'(x_i) = 0$$

for each $i = 0, 1, \ldots, n$.

Thus, $D(x)$ has roots of multiplicity at least 2 at each of the $n+1$ distinct points $x_0, \ldots, x_n$. It follows that

$$D(x) = \prod_{i=0}^{n} (x - x_i)^2 \, Q(x),$$

where $Q(x)$ is a polynomial. The factor $\prod_{i=0}^{n} (x - x_i)^2$ has degree $2(n+1) = 2n+2$. Since $\deg(D) \leq 2n+1$, the only possibility is $Q(x) \equiv 0$, which implies $D(x) \equiv 0$. Hence, $H_{2n+1}(x) = P(x)$, so $H_{2n+1}(x)$ is the unique such polynomial. As its degree is at most $2n+1$, it is also the one of least degree.

(b)

Assume $f \in C^{2n+2}[a, b]$ with $x_0, \ldots, x_n \in [a, b]$ distinct and $x \in [a, b]$ (with $x \neq x_i$ for all i). Let $\omega(t) = \prod_{i=0}^{n} (t - x_i)^2$, a polynomial of degree $2n+2$. Define

$$g(t) = f(t) - H_{2n+1}(t) - \lambda \, \omega(t),$$

where

$$\lambda = \frac{f(x) - H_{2n+1}(x)}{\omega(x)}.$$

Then $g(x) = 0$. Moreover, for each $i = 0, \ldots, n$,

$$g(x_i) = f(x_i) - H_{2n+1}(x_i) = 0, \quad g'(x_i) = f'(x_i) - H'_{2n+1}(x_i) = 0,$$

and $\omega(t)$ (hence $\lambda \, \omega(t)$) vanishes to order 2 at each x_i. Thus, $g(t)$ vanishes to order at least 2 at each of the $n+1$ points $x_0, \ldots, x_n$ (total multiplicity at least $2(n+1) = 2n+2$) and has an additional zero at x (multiplicity at least 1), for a total of at least $2n+3$ zeros (counting multiplicity).

By repeated application of Rolle's theorem, $g'(t)$ has at least $2n+2$ zeros (counting multiplicity), $g''(t)$ has at least $2n+1$ zeros, ..., and $g^{(2n+2)}(t)$ has at least one zero in (a, b). Now compute

$$g^{(2n+2)}(t) = f^{(2n+2)}(t) - \lambda \, \omega^{(2n+2)}(t).$$

Since $\omega(t)$ is monic of degree $2n+2$, its leading term is t^{2n+2}, so $\omega^{(2n+2)}(t) = (2n+2)!$. Thus,

$$g^{(2n+2)}(t) = f^{(2n+2)}(t) - \lambda \, (2n+2)!.$$

This equals zero at some $\xi = \xi(x) \in (a, b)$, so

$$\lambda = \frac{f^{(2n+2)}(\xi)}{(2n+2)!}.$$

Substituting back,

$$f(x) - H_{2n+1}(x) = \lambda \, \omega(x) = \frac{f^{(2n+2)}(\xi(x))}{(2n+2)!} \prod_{i=0}^{n} (x - x_i)^2,$$

as required.

Problem 2: Problems from the textbook, chapter 4.1

2a

The standard **three-point formulas** for equally spaced nodes (h) are used:

- **Three-point endpoint formula (forward difference)** at a left endpoint:

$$f'(x_0) \approx \frac{-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)}{2h}.$$

- **Three-point midpoint (centered) formula**:

$$f'(x_1) \approx \frac{f(x_1+h) - f(x_1-h)}{2h}.$$

- **Three-point endpoint formula (backward difference)** at a right endpoint:

$$f'(x_2) \approx \frac{f(x_0) - 4f(x_1) + 3f(x_2)}{2h},$$

which is the same endpoint formula but “from the right side”.

$$\begin{aligned}
f'(-0.3) &\approx \frac{-3f(-0.3) + 4f(-0.2) - f(-0.1)}{2(0.1)} \\
&= \frac{-3(1.9507) + 4(2.0421) - 2.0601}{0.2} \\
&= \frac{-5.8521 + 8.1684 - 2.0601}{0.2} = \frac{0.2562}{0.2} = 1.2810.
\end{aligned}$$

$$\begin{aligned}
f'(-0.2) &\approx \frac{f(-0.1) - f(-0.3)}{2(0.1)} \\
&= \frac{2.0601 - 1.9507}{0.2} = \frac{0.1094}{0.2} = 0.5470.
\end{aligned}$$

$$\begin{aligned}
f'(-0.1) &\approx \frac{f(-0.3) - 4f(-0.2) + 3f(-0.1)}{2(0.1)} \\
&= \frac{1.9507 - 4(2.0421) + 3(2.0601)}{0.2} \\
&= \frac{1.9507 - 8.1684 + 6.1803}{0.2} = \frac{-0.0374}{0.2} = -0.1870.
\end{aligned}$$

The completed table is (to four decimals):

x	f(x)	f'(x) approx
-0.3	1.9507	1.2810
-0.2	2.0421	0.5470
-0.1	2.0601	-0.1870

4a

The *true* function $f(x) = 2\cos(2x) - x$ is used to get exact derivatives and compare.

$$f(x) = 2\cos(2x) - x \quad \Longrightarrow \quad f'(x) = -4\sin(2x) - 1.$$

Compute at each point:

- At $x = -0.3$:

$$f'(-0.3) = -4\sin(-0.6) - 1 = 4\sin(0.6) - 1 \approx 1.25857.$$

- At $x = -0.2$:

$$f'(-0.2) = -4\sin(-0.4) - 1 = 4\sin(0.4) - 1 \approx 0.55767.$$

- At $x = -0.1$:

$$f'(-0.1) = -4\sin(-0.2) - 1 = 4\sin(0.2) - 1 \approx -0.20532.$$

Actual errors = (approx – true):

- At $x = -0.3$:

$$E(-0.3) = 1.2810 - 1.25857 \approx 0.02243, \quad |E(-0.3)| \approx 2.24 \times 10^{-2}.$$

- At $x = -0.2$:

$$E(-0.2) = 0.5470 - 0.55767 \approx -0.01067, \quad |E(-0.2)| \approx 1.07 \times 10^{-2}.$$

- At $x = -0.1$:

$$E(-0.1) = -0.1870 - (-0.20532) \approx 0.01832, \quad |E(-0.1)| \approx 1.83 \times 10^{-2}.$$

From the theory for **three-point endpoint** and **three-point midpoint** formulas with step h:

- Endpoint formula:

$$f'(x_0) = \frac{-3f(x_0) + 4f(x_0+h) - f(x_0+2h)}{2h} + \frac{h^2}{3} f^{(3)}(\xi_0),$$

so

$$|E_{\text{end}}| \le \frac{h^2}{3} \max |f^{(3)}(x)|.$$

- Midpoint (centered) formula:

$$f'(x_1) = \frac{f(x_1+h) - f(x_1-h)}{2h} - \frac{h^2}{6} f^{(3)}(\xi_1),$$

so

$$|E_{\text{mid}}| \le \frac{h^2}{6} \max |f^{(3)}(x)|.$$

For $f(x)$:

$$f'(x) = -4\sin(2x) - 1, \quad f''(x) = -8\cos(2x), \quad f^{(3)}(x) = 16\sin(2x).$$

All formulas use points between -0.3 and -0.1, so $f^{(3)}$ is bounded on that interval:
$x \in [-0.3, -0.1] \Rightarrow 2x \in [-0.6, -0.2]$.
On $[-0.6, -0.2]$, $|\sin(2x)|$ is maximized at $2x = -0.6$, so

$$\max_{[-0.3,-0.1]} |f^{(3)}(x)| = 16 \max |\sin(2x)| \approx 16|\sin(-0.6)| \approx 9.03.$$

Take $M \approx 9.03$, $h = 0.1$, $h^2 = 0.01$:

- **Endpoint error bound** (for $x = -0.3$ and $x = -0.1$):

$$|E_{\text{end}}| \le \frac{h^2}{3} M = \frac{0.01}{3} \cdot 9.03 \approx 0.0301.$$

- **Midpoint error bound** (for $x = -0.2$):

$$|E_{\text{mid}}| \le \frac{h^2}{6} M = \frac{0.01}{6} \cdot 9.03 \approx 0.0151.$$

Compare with the **actual** errors:

- $|E(-0.3)| \approx 0.0224 \le 0.0301$.
- $|E(-0.2)| \approx 0.0107 \le 0.0151$.
- $|E(-0.1)| \approx 0.0183 \le 0.0301$.

The numerical errors from the forward/backward (three-point) formulas are all comfortably within the theoretical error bounds predicted by the $O(h^2)$ truncation error terms.

5d

Formulas (with $h = 0.1$):

- Left endpoint x_0:

$$f'(x_0) \approx \frac{-3f(x_0) + 4f(x_0+h) - f(x_0+2h)}{2h}$$

- Interior (midpoint) at $x_0 + h$:

$$f'(x_0+h) \approx \frac{f(x_0+2h) - f(x_0)}{2h}$$

- Interior (midpoint) at $x_0 + 2h$:

$$f'(x_0+2h) \approx \frac{f(x_0+3h) - f(x_0+h)}{2h}$$

- Right endpoint x_3:

$$f'(x_3) \approx \frac{3f(x_3) - 4f(x_3-h) + f(x_3-2h)}{2h}$$

Plugging in the values:

- $f(2.0) = 3.6887983$
- $f(2.1) = 3.6905701$
- $f(2.2) = 3.6688192$
- $f(2.3) = 3.6245909$
- $h = 0.1, 2h = 0.2$

1. **At** $x = 2.0$ (left endpoint):

$$
\begin{aligned}
f'(2.0) &\approx \frac{-3f(2.0) + 4f(2.1) - f(2.2)}{0.2} \\
&= \frac{-3(3.6887983) + 4(3.6905701) - 3.6688192}{0.2} \\
&\approx 0.1353315
\end{aligned}
$$

2. **At** $x = 2.1$ (midpoint):

$$
\begin{aligned}
f'(2.1) &\approx \frac{f(2.2) - f(2.0)}{0.2} \\
&= \frac{3.6688192 - 3.6887983}{0.2} \\
&\approx -0.0998955
\end{aligned}
$$

3. **At** $x = 2.2$ (midpoint):

$$
\begin{aligned}
f'(2.2) &\approx \frac{f(2.3) - f(2.1)}{0.2} \\
&= \frac{3.6245909 - 3.6905701}{0.2} \\
&\approx -0.3298960
\end{aligned}
$$

4. **At** $x = 2.3$ (right endpoint):

$$
\begin{aligned}
f'(2.3) &\approx \frac{3f(2.3) - 4f(2.2) + f(2.1)}{0.2} \\
&= \frac{3(3.6245909) - 4(3.6688192) + 3.6905701}{0.2} \\
&\approx -0.5546700
\end{aligned}
$$

Rounding to 6 decimal places, the completed table is:

x	f(x)	f'(x)
2.0	3.6887983	0.135332
2.1	3.6905701	-0.099896
2.2	3.6688192	-0.329896
2.3	3.6245909	-0.554670

24

All functions are expanded around x_0. For brevity, let $f^{(k)}(x_0) = f_k$.

$$
\begin{aligned}
f(x_0 - h) &= f_0 - hf_1 + \frac{h^2}{2}f_2 - \frac{h^3}{6}f_3 + \frac{h^4}{24}f_4 - \frac{h^5}{120}f_5 + \cdots, \\
f(x_0) &= f_0, \\
f(x_0 + h) &= f_0 + hf_1 + \frac{h^2}{2}f_2 + \frac{h^3}{6}f_3 + \frac{h^4}{24}f_4 + \frac{h^5}{120}f_5 + \cdots, \\
f(x_0 + 2h) &= f_0 + 2hf_1 + 2h^2f_2 + \frac{4}{3}h^3f_3 + \frac{2}{3}h^4f_4 + \frac{4}{15}h^5f_5 + \cdots, \\
f(x_0 + 3h) &= f_0 + 3hf_1 + \frac{9}{2}h^2f_2 + \frac{9}{2}h^3f_3 + \frac{27}{8}h^4f_4 + \frac{81}{40}h^5f_5 + \cdots.
\end{aligned}
$$

Form the linear combination

$$
A = af(x_0 - h) + bf(x_0) + cf(x_0 + h) + df(x_0 + 2h) + ef(x_0 + 3h).
$$

Substitute the expansions and collect by derivatives:

- Coefficient of f_0:

$$
a + b + c + d + e
$$

- Coefficient of hf_1:

$$
-a + c + 2d + 3e
$$

- Coefficient of h^2f_2:

$$
\frac{a}{2} + \frac{c}{2} + 2d + \frac{9}{2}e
$$

- Coefficient of h^3f_3:

$$
-\frac{a}{6} + \frac{c}{6} + \frac{4}{3}d + \frac{9}{2}e
$$

- Coefficient of h^4f_4:

$$
\frac{a}{24} + \frac{c}{24} + \frac{2}{3}d + \frac{27}{8}e
$$

- Coefficient of $h^5 f_5$:

$$-\frac{a}{120} + \frac{c}{120} + \frac{4}{15}d + \frac{81}{40}e$$

Then

$$\frac{A}{h} = (\text{coeff}_1)f_1 + (\text{coeff}_2)hf_2 + (\text{coeff}_3)h^2 f_3 + (\text{coeff}_4)h^3 f_4 + (\text{coeff}_5)h^4 f_5 + \cdots$$

and this must approximate $f_1 = f'(x_0)$ with error $O(h^4)$.

The following conditions must be met:

1. No f_0 term:

$$a + b + c + d + e = 0$$

2. Correct coefficient on f_1:

$$-a + c + 2d + 3e = 1$$

3. Kill the f_2 term:

$$\frac{a}{2} + \frac{c}{2} + 2d + \frac{9}{2}e = 0$$

4. Kill the f_3 term:

$$-\frac{a}{6} + \frac{c}{6} + \frac{4}{3}d + \frac{9}{2}e = 0$$

5. Kill the f_4 term:

$$\frac{a}{24} + \frac{c}{24} + \frac{2}{3}d + \frac{27}{8}e = 0$$

The solution to this 5×5 linear system (for a, b, c, d, e) is

$$a = -\frac{1}{4}, \quad b = -\frac{5}{6}, \quad c = \frac{3}{2}, \quad d = -\frac{1}{2}, \quad e = \frac{1}{12}.$$

So

$$\begin{aligned} f'(x_0) &\approx \frac{1}{h}\Big[-\tfrac{1}{4}f(x_0 - h) - \tfrac{5}{6}f(x_0) + \tfrac{3}{2}f(x_0 + h) - \tfrac{1}{2}f(x_0 + 2h) + \tfrac{1}{12}f(x_0 + 3h)\Big] \\ &= \frac{1}{12h}\Big[-3f(x_0 - h) - 10f(x_0) + 18f(x_0 + h) - 6f(x_0 + 2h) + f(x_0 + 3h)\Big]. \end{aligned}$$

Keeping the next term in the Taylor series yields the truncation error:

$$f'(x_0) = \frac{1}{12h}\big[-3f(x_0 - h) - 10f(x_0) + 18f(x_0 + h) - 6f(x_0 + 2h) + f(x_0 + 3h)\big] + \frac{h^4}{20}f^{(5)}(\xi),$$

for some ξ between $x_0 - h$ and $x_0 + 3h$.

Hence, this is an $O(h^4)$ five-point formula using exactly the points
$x_0 - h, x_0, x_0 + h, x_0 + 2h, x_0 + 3h$.

Problem 3: Problems from the textbook, chapter 4.2

1a

For the derivative at $x_0 = 1$, with step h, the three-point midpoint formula is

$$N_1(h) \equiv f'(1) \approx \frac{f(1 + h) - f(1 - h)}{2h}.$$

We need $N_1(0.4), N_1(0.2), N_1(0.1)$.

Using $f(x) = \ln x$:

- For $h = 0.4$: points (0.6, 1.4)

$$N_1(0.4) = \frac{\ln(1.4) - \ln(0.6)}{0.8} \approx 1.0591223.$$

- For $h = 0.2$: points (0.8, 1.2)

$$N_1(0.2) = \frac{\ln(1.2) - \ln(0.8)}{0.4} \approx 1.0136628.$$

- For $h = 0.1$: points (0.9, 1.1)

$$N_1(0.1) = \frac{\ln(1.1) - \ln(0.9)}{0.2} \approx 1.0033535.$$

For an even-power error expansion

$$M - N_1(h) = K_1 h^2 + K_2 h^4 + \cdots,$$

the extrapolated formula is

$$N_2(h) = N_1\left(\frac{h}{2}\right) + \frac{N_1\left(\frac{h}{2}\right) - N_1(h)}{3}.$$

- For $h = 0.4$:

$$\begin{aligned} N_2(0.4) &= N_1(0.2) + \frac{N_1(0.2) - N_1(0.4)}{3} \\ &\approx 1.0136628 + \frac{1.0136628 - 1.0591223}{3} \\ &\approx 0.9985096. \end{aligned}$$

- For $h = 0.2$:

$$\begin{aligned} N_2(0.2) &= N_1(0.1) + \frac{N_1(0.1) - N_1(0.2)}{3} \\ &\approx 1.0033535 + \frac{1.0033535 - 1.0136628}{3} \\ &\approx 0.9999170. \end{aligned}$$

The next level (again for even powers) is

$$N_3(h) = N_2\left(\frac{h}{2}\right) + \frac{N_2\left(\frac{h}{2}\right) - N_2(h)}{15},$$

since $4^2 - 1 = 15$.

For $h = 0.4$:

$$\begin{aligned} N_3(0.4) &= N_2(0.2) + \frac{N_2(0.2) - N_2(0.4)}{15} \\ &\approx 0.9999170 + \frac{0.9999170 - 0.9985096}{15} \\ &\approx 0.9999170 + 0.0000939 \\ &\approx 1.0000109. \end{aligned}$$

The exact derivative is

$$f'(x) = \frac{1}{x} \quad \Rightarrow \quad f'(1) = 1.$$

The extrapolated value is

$$N_3(0.4) \approx 1.0000109$$

which differs from the exact value 1 by only about 1.1×10^{-5}.

5

Column 1 consists of the given $N_1(h/2^k)$ for $k = 0, 1, 2, 3$. Columns 2–4 are then built with the recursion above.

step	N_1
h	1.570796
$h/2$	1.896119
$h/4$	1.974232
$h/8$	1.993570

Using

$$N_2(h) = N_1(h/2) + \frac{N_1(h/2) - N_1(h)}{3},$$

and similarly for smaller steps:

$$\begin{gathered} N_2(h) \approx 2.004560, \\ N_2(h/2) \approx 2.0002696667, \\ N_2(h/4) \approx 2.000016. \end{gathered}$$

$$N_3(h) = N_2(h/2) + \frac{N_2(h/2) - N_2(h)}{4^2 - 1} = N_2(h/2) + \frac{N_2(h/2) - N_2(h)}{15},$$

$$\begin{gathered} N_3(h) \approx 1.9999836444, \\ N_3(h/2) \approx 1.9999990889. \end{gathered}$$

$$N_4(h) = N_3(h/2) + \frac{N_3(h/2) - N_3(h)}{4^3 - 1} = N_3(h/2) + \frac{N_3(h/2) - N_3(h)}{63}.$$

$$\boxed{N_4(h) \approx 1.999999334038801}.$$

The exact integral is $\int_0^\pi \sin x\, dx = 2$, so the error is about 6.7×10^{-7}.

The full Richardson table (numbers rounded) is:

step	N_1	N_2	N_3	N_4
h	1.570796	2.004560	1.9999836	1.9999993
$h/2$	1.896119	2.0002697	1.9999991	
$h/4$	1.974232	2.000016		
$h/8$	1.993570			

Assuming the given values correspond to $N_2(h/2^k)$ with an error expansion $M = N_2(h) + K_2h^2 + K_4h^4 + K_6h^6 + K_8h^8 + O(h^{10})$, the value of $N_5(h)$ is determined.

If the first column is treated as N_2 instead of N_1, the **numerical extrapolation process is identical**; only the names of the columns shift:

- Column 1: $N_2(h/2^k)$
- Column 2: $N_3(h/2^k)$
- Column 3: $N_4(h/2^k)$
- Column 4: $N_5(h)$

The value at the bottom of the 4th column is now labeled $N_5(h)$, and its numerical value is the same as the previously computed $N_4(h)$:

$$N_5(h) \approx 1.999999334038801.$$

Both $N_4(h)$ in the first interpretation and $N_5(h)$ in the second interpretation approximate the exact integral to about 6–7 correct decimal places.

9

The given expansion is $M = N(h) + K_1h + K_2h^2 + K_3h^3 + \cdots$. To find an $O(h^3)$ approximation using $N(h)$, $N(h/3)$, and $N(h/9)$, form the linear combination $\frac{1}{16}N(h) - \frac{3}{4}N\left(\frac{h}{3}\right) + \frac{27}{16}N\left(\frac{h}{9}\right)$, or equivalently $\frac{1}{16}\left[N(h) - 12N\left(\frac{h}{3}\right) + 27N\left(\frac{h}{9}\right)\right]$.

To derive this, substitute the expansions:

$$\begin{aligned}
M &= N(h) + K_1h + K_2h^2 + K_3h^3 + O(h^4), \\
M &= N\left(\frac{h}{3}\right) + K_1\frac{h}{3} + K_2\left(\frac{h}{3}\right)^2 + K_3\left(\frac{h}{3}\right)^3 + O(h^4), \\
M &= N\left(\frac{h}{9}\right) + K_1\frac{h}{9} + K_2\left(\frac{h}{9}\right)^2 + K_3\left(\frac{h}{9}\right)^3 + O(h^4).
\end{aligned}$$

Consider the combination $N = aN(h) + bN\left(\frac{h}{3}\right) + cN\left(\frac{h}{9}\right)$. Then

$$N = (a+b+c)M - K_1h\left(a + \frac{b}{3} + \frac{c}{9}\right) - K_2h^2\left(a + \frac{b}{9} + \frac{c}{81}\right) - K_3h^3\left(a + \frac{b}{27} + \frac{c}{729}\right) + O(h^4).$$

Solve the system

$$\begin{aligned}
a + b + c &= 1, \\
a + \frac{b}{3} + \frac{c}{9} &= 0, \\
a + \frac{b}{9} + \frac{c}{81} &= 0.
\end{aligned}$$

Subtracting the second equation from the first gives $\frac{2}{3}b + \frac{8}{9}c = 1$. Subtracting the third from the second gives $\frac{2}{9}b + \frac{8}{81}c = 0$, so $b = -\frac{4}{9}c$. Substituting yields $c = \frac{27}{16}$, $b = -\frac{3}{4}$, and $a = \frac{1}{16}$. The error is then $O(h^3)$.

10

To produce an $O(h^6)$ approximation to M, solve the system of equations for coefficients a, b, and c such that $a + b + c = 1$, $a + \frac{1}{9}b + \frac{1}{81}c = 0$, and $a + \frac{1}{81}b + \frac{1}{6561}c = 0$. The solution is $a = \frac{1}{640}$, $b = -\frac{9}{64}$, and $c = \frac{729}{640}$.

The approximation is then $aN(h) + bN\left(\frac{h}{3}\right) + cN\left(\frac{h}{9}\right)$, which simplifies to $\frac{N(h) - 90N\left(\frac{h}{3}\right) + 729N\left(\frac{h}{9}\right)}{640}$.

Problem 4: Derivation of finite difference formula

A linear combination of $N(h), N(h/3), N(h/9)$ is sought, whose error is $O(h^3)$.

The given error expansion is

$$M = N(h) + K_1 h + K_2 h^2 + K_3 h^3 + \cdots$$

for all $h > 0$. For smaller step sizes:

$$M = N\left(\tfrac{h}{3}\right) + K_1 \tfrac{h}{3} + K_2 \tfrac{h^2}{9} + K_3 \tfrac{h^3}{27} + \cdots,$$
$$M = N\left(\tfrac{h}{9}\right) + K_1 \tfrac{h}{9} + K_2 \tfrac{h^2}{81} + K_3 \tfrac{h^3}{729} + \cdots.$$

These can be rewritten as expansions for $N(\cdot)$:

$$N(h) = M - K_1 h - K_2 h^2 - K_3 h^3 + \cdots,$$
$$N\left(\tfrac{h}{3}\right) = M - K_1 \tfrac{h}{3} - K_2 \tfrac{h^2}{9} - K_3 \tfrac{h^3}{27} + \cdots,$$
$$N\left(\tfrac{h}{9}\right) = M - K_1 \tfrac{h}{9} - K_2 \tfrac{h^2}{81} - K_3 \tfrac{h^3}{729} + \cdots.$$

An approximation of the form is considered:

$$N_*(h) = aN(h) + bN\left(\tfrac{h}{3}\right) + cN\left(\tfrac{h}{9}\right).$$

Substituting the series:

$$\begin{aligned} N_*(h) = a\big[M - K_1 h - K_2 h^2 - K_3 h^3 - \cdots\big] \\ + b\big[M - K_1 \tfrac{h}{3} - K_2 \tfrac{h^2}{9} - K_3 \tfrac{h^3}{27} - \cdots\big] \\ + c\big[M - K_1 \tfrac{h}{9} - K_2 \tfrac{h^2}{81} - K_3 \tfrac{h^3}{729} - \cdots\big]. \end{aligned}$$

Grouping terms:

- Coefficient of M: $a + b + c$.
- Coefficient of $K_1 h$: $-(a + \frac{b}{3} + \frac{c}{9})$.
- Coefficient of $K_2 h^2$: $-(a + \frac{b}{9} + \frac{c}{81})$.

To have

$$N_*(h) = M + O(h^3),$$

the following conditions are required:

$$\begin{cases} a + b + c = 1 & \text{(correct constant term)} \\ a + \dfrac{b}{3} + \dfrac{c}{9} = 0 & \text{(kill the } h \text{ term)} \\ a + \dfrac{b}{9} + \dfrac{c}{81} = 0 & \text{(kill the } h^2 \text{ term).} \end{cases}$$

Solving this system gives

$$a = \frac{1}{16}, \quad b = -\frac{3}{4}, \quad c = \frac{27}{16}.$$

An $O(h^3)$ approximation to M is

$$\begin{aligned} M \approx N_*(h) &= \frac{1}{16} N(h) - \frac{3}{4} N\left(\frac{h}{3}\right) + \frac{27}{16} N\left(\frac{h}{9}\right) \\ &= \frac{N(h) - 12N(h/3) + 27N(h/9)}{16}. \end{aligned}$$

The remaining leading error term is proportional to h^3.

Problem 5: Use of fdstencil

(a)

To derive a fourth-order accurate finite difference approximation to $f''(x)$ using the 5 equally spaced points $x - 2h$, $x - h$, x, $x + h$, and $x + 2h$, apply Taylor expansions of $f(x + kh)$ for $k = -2, -1, 0, 1, 2$ around x:

$$f(x + kh) = \sum_{m=0}^{\infty} \frac{(kh)^m}{m!} f^{(m)}(x).$$

Substitute into the linear combination $\sum_{k=-2}^{2} c_k f(x + kh)$ and collect terms to obtain

$$\sum_{k=-2}^{2} c_k f(x + kh) = \sum_{m=0}^{\infty} \frac{h^m}{m!} \left(\sum_{k=-2}^{2} c_k k^m \right) f^{(m)}(x).$$

For the approximation to satisfy $f''(x) = \sum_{k=-2}^{2} c_k f(x + kh) + O(h^4)$, the inner sums must satisfy $\sum_{k=-2}^{2} c_k k^m = 0$ for $m = 0, 1, 3, 4$ and $\sum_{k=-2}^{2} c_k k^2 = 2$ for $m = 2$ (ensuring the $h^2/2!$ coefficient yields exactly $f''(x)$, with the error starting at the $O(h^4)$ term from $m = 6$ due to symmetry canceling the $m = 5$ term). This yields the 5×5 linear system $V\mathbf{c} = \mathbf{b}$, where $\mathbf{c} = [c_{-2}, c_{-1}, c_0, c_1, c_2]^T$,

$$V = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 & 2 \\ 4 & 1 & 0 & 1 & 4 \\ -8 & -1 & 0 & 1 & 8 \\ 16 & 1 & 0 & 1 & 16 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ 2 \\ 0 \\ 0 \end{pmatrix}.$$

(Note: These **c** are such that the full approximation is actually $f''(x) \approx h^{-2} \sum_{k=-2}^{2} c_k f(x+kh) + O(h^4)$, as the h^{-2} scaling is standard for second derivatives; the system above determines the $O(1)$ coefficients before scaling.)

(b)

The coefficients satisfying the system in part (a) are

$$c_{-2} = -\frac{1}{12}, \quad c_{-1} = \frac{4}{3}, \quad c_0 = -\frac{5}{2}, \quad c_1 = \frac{4}{3}, \quad c_2 = -\frac{1}{12}.$$

These match the output of `fdstencil(2, -2:2)`, which prints:

```
The derivative f^(2) of u at x0 is approximated by

 1/h^2 * [
 -0.08333333333333322 * f(x0-2*h) +
  1.3333333333333333 * f(x0-h) +
 -2.5000000000000000 * f(x0) +
  1.3333333333333333 * f(x0+h) +
 -0.08333333333333322 * f(x0+2*h) ]

For smooth f,
 Error = 0 * h^3*f^(5) + -0.011111111111111112 * h^4*f^(6) + ...
```

To verify, compute $\sum c_k k^m$:

- $m = 0$: $-1/12 + 4/3 - 5/2 + 4/3 - 1/12 = 0$
- $m = 1$: $2(-1/12) + (-1)(4/3) + 1(4/3) + 2(-1/12) = 0$ (odd powers vanish by symmetry)
- $m = 2$: $4(-1/12) + 1(4/3) + 0 + 1(4/3) + 4(-1/12) = 2$
- $m = 3$: 0 (by symmetry)
- $m = 4$: $16(-1/12) + 1(4/3) + 0 + 1(4/3) + 16(-1/12) = 0$

These satisfy the system exactly.

(c)

For $f(x) = \sin(2x)$, $f''(x) = -4\sin(2x)$, so the exact value is $f''(1) = -4\sin(2) \approx -3.63719$. The approximation is

$$f''(1) \approx \frac{1}{h^2} \sum_{k=-2}^{2} c_k \sin(2(1+kh)),$$

with coefficients from part (b). The leading error term from `fdstencil` is $-\frac{1}{90} h^4 f^{(6)}(\xi)$. For this f, $f^{(6)}(x) = -64\sin(2x)$, so at $x = 1$, the predicted error is $\approx +\frac{64}{90}\sin(2)\, h^4 \approx +0.6467 h^4$.

The array `hvals = logspace(-1, -4, 13)` gives $h = 10^{-1}, 10^{-1.25}, \ldots, 10^{-4}$. Below is a table of the absolute error $|\hat{f}''(1) - f''(1)|$ vs. h for 5 representative values (larger h show the $O(h^4)$ convergence; smaller h show degradation due to subtractive cancellation in floating-point arithmetic when forming the linear combination). The predicted error uses the leading term evaluated at $x = 1$.

h	Absolute Error	Predicted Error
1.00×10^{-1}	6.44×10^{-5}	6.47×10^{-5}
1.78×10^{-2}	6.47×10^{-8}	6.47×10^{-8}
3.16×10^{-3}	5.20×10^{-11}	6.47×10^{-11}
5.62×10^{-4}	4.82×10^{-10}	6.47×10^{-14}
1.00×10^{-4}	3.38×10^{-8}	6.47×10^{-17}